

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
    c.state,
    COUNT(DISTINCT o.customer_id) AS customers_with_orders
FROM order_t o
    JOIN customer_t c
    ON o.customer_id = c.customer_id
GROUP BY c.state
ORDER BY customers_with_orders DESC;
```

Output:

state	customers_with_orders
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35
Colorado	33
Ohio	33
Alabama	29
Washington	28
Arizona	26
Illinois	25
Pennsylvania	25
Virginia	24
North Dakota	2
Mississippi	2
Wyoming	1
Vermont	1
Maine	1

Observations and Insights:

- California (97), Texas (97), and Florida (86) lead the top 3 of customers who placed orders, mid-tier New York (69), District of Columbia(35), and Colorado (33), low performing North Dakota (2), Mississippi (2),Wyoming (1), Vermont (1) and Maine (1)
- There is a sharp drop from states with 97 customers to states with 1-2 customers.
- Focus marketing and logistics on top states, Expansion opportunities for low performing states.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.vehicle_maker,
    COUNT(o.order_id) as Total_orders
FROM order_t o
    JOIN product_t p
        ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY Total_orders DESC
LIMIT 5;
```

Output:

vehicle_maker	Total_orders
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

#	Time	Action	Message
44	21:01:54	SELECT p.vehicle_maker, COUNT(o.order_id) as Total_orders FROM order_t o JOIN product_t p ON o.product_id = p.product_id GROUP BY p.veh...	5 row(s) returned

Observations and Insights:

- The top 5 preferred vehicle makers are Chevrolet (83), Ford (63), Toyota (52), Pontiac and Dodge both with (50).
- Maintain partnerships with these most preferred vehicle maker, exclusive offers or loyalty programs could further boost sales.
- Investigate brand specific satisfaction and feedback, this can reveal whether high order volume also translates into customer satisfaction and retention.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state,
       vehicle_maker,
       total_orders
FROM (
    SELECT c.state,
           p.vehicle_maker,
           COUNT(o.order_id) AS total_orders,
           RANK() OVER(PARTITION BY c.state ORDER BY COUNT(o.order_id) DESC) AS rnk
    FROM order_t o
    JOIN customer_t c
        ON o.customer_id = c.customer_id
    JOIN product_t p
        ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
) sub
WHERE rnk = 1
ORDER BY state;
```

Output:

state	vehicle_maker	total_orders
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Cadillac	3
Arizona	Pontiac	3
Arkansas	GMC	1
Arkansas	Suzuki	1
Arkansas	Volkswagen	1
Arkansas	Chevrolet	1
Arkansas	Pontiac	1
Arkansas	Mitsubishi	1
California	Audi	6
California	Chevrolet	6
California	Nissan	6
California	Dodge	6
California	Ford	6
Colorado	Chevrolet	5

Result 7 ×

Output

Action Output

#	Time	Action	Message
8	19:50:04	SELECT state, vehicle_maker, total_orders FROM (SELECT c.state, p.vehicle_maker, COUNT(o.order_id) AS total_orders, ...	143 row(s) returned

Observations and Insights:

- Chevrolet appears multiple times (e.g. Alaska, California, Texas, Ohio, Colorado) show nationwide brand strength.
- Dodge is preferred in Alabama, possibly reflecting demand for trucks or utility vehicles in that region.
- States with fragmented or minimal brand orders (e.g. Arkansas, Montana, Vermont) indicate underdeveloped sales channels.
- States like California exhibit a competitive automotive market with no clear leader, suggesting diverse consumer preferences.

Question 4: Find the overall average rating given by the customers.
What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SELECT
  CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter,
  AVG(
    CASE customer_feedback
      WHEN 'Very Bad' THEN 1
      WHEN 'Bad' THEN 2
      WHEN 'Okay' THEN 3
      WHEN 'Good' THEN 4
      WHEN 'Very Good' THEN 5
    END
  ) AS Overall_Rating
FROM order_t
GROUP BY year_quarter
ORDER BY year_quarter;
```

Output:

year_quarter	Overall_Rating
2018-Q1	3.5548
2018-Q2	3.3550
2018-Q3	2.9563
2018-Q4	2.3970

#	Time	Action	Message
24	10:11:43	SELECT CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter, AVG(CASE customer_feedback WHEN 'Very...	4 row(s) returned

Observations and Insights:

- Declining Trend in customer satisfaction, the average drop from 3.55 in Q1 to 2.40 in Q4, this suggests growing dissatisfaction over time.
- Sharp drop in the last two quarters, between Q3 (2.96) and Q4 (2.40) a fall of more than 18%.
- Investigate root causes of decline, identify if low ratings are concentrated among specific brands or regions. Monitor quarter over quarter satisfaction

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
  CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter,
  ROUND(100.0 * SUM(CASE WHEN customer_feedback IN ('Good','Very Good') THEN 1 ELSE 0 END) / COUNT(*), 2) AS satisfied_pct,
  ROUND(100.0 * SUM(CASE WHEN customer_feedback IN ('Okay') THEN 1 ELSE 0 END) / COUNT(*), 2) AS neutral_pct,
  ROUND(100.0 * SUM(CASE WHEN customer_feedback IN ('Bad','Very Bad') THEN 1 ELSE 0 END) / COUNT(*), 2) AS dissatisfied_pct
FROM order_t
GROUP BY year_quarter
ORDER BY year_quarter;
```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
year_quarter	satisfied_pct	neutral_pct	dissatisfied_pct
2018-Q1	58.71	19.03	22.26
2018-Q2	50.76	20.23	29.01
2018-Q3	37.55	21.83	40.61
2018-Q4	20.10	20.10	59.80

#	Time	Action	Message
9	11:39:38	SELECT CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter, customer_feedback, ROUND(100.0 * COUNT(*) O...	20 row(s) returned

Observations and Insights:

- Strong decline in customer satisfaction, customer satisfied decreased from **58.7% in Q1** to only **20.1% in Q4** — a **drop of nearly 66%** in one year.
- Neutral feedback remains stable between **19-21%**, suggesting that the major shift came from satisfied customers turning into dissatisfied ones.
- The trend shows a growing negative perception, highlighting the need for immediate quality and service improvements.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
WITH orders_per_quarter AS (
  SELECT
    CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter,
    COUNT(order_id) AS total_orders
  FROM order_t
  GROUP BY CONCAT(YEAR(order_date), '-Q', QUARTER(order_date))
)
SELECT
  year_quarter,
  total_orders,
  ROUND(
    ((total_orders - LAG(total_orders) OVER (ORDER BY year_quarter))
     / LAG(total_orders) OVER (ORDER BY year_quarter)) * 100, 2
  ) AS qoq_change_pct
FROM orders_per_quarter
ORDER BY year_quarter;
```

Output:

year_quarter	total_orders	qoq_change_pct
2018-Q1	310	NULL
2018-Q2	262	-15.48
2018-Q3	229	-12.60
2018-Q4	199	-13.10

Result 17 x

Output

Action Output

#	Time	Action	Message
21	14:32:12	WITH orders_per_quarter AS (SELECT CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter, COUNT(order_id) A...	4 row(s) returned

Observations and Insights:

- Strong decline in placed orders, decreased from **310 in Q1** to only **199 in Q4**.
- Quarter-over-Quarter (QoQ) Decline
 - Q1 → Q2: -15.48% drop
 - Q2 → Q3: -12.6% drop
 - Q3 → Q4: -13.1% drop
- Customer retention problem, the steady decline suggests customers may not be placing repeat orders due to poor experiences, operational or logistics factors, delays in shipping, product quality or pricing strategy could have affected sales post Q1.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

-- Total net revenue --

```
SELECT ROUND(SUM(quantity * vehicle_price * (1 - discount/100)), 2) AS net_revenue
FROM order_t;
```

-- Quarterly revenue and QoQ % --

```
WITH revenue_by_quarter AS (
  SELECT
    CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter,
    ROUND(SUM(quantity * vehicle_price * (1 - discount/100)), 2) AS net_revenue
  FROM order_t
  GROUP BY year_quarter
)
SELECT
  year_quarter,
  net_revenue,
  LAG(net_revenue) OVER (ORDER BY year_quarter) AS prev_revenue,
  ROUND(100.0 * (net_revenue - LAG(net_revenue) OVER (ORDER BY year_quarter))
    / NULLIF(LAG(net_revenue) OVER (ORDER BY year_quarter), 0), 2) AS qoq_change_pct
FROM revenue_by_quarter
ORDER BY year_quarter;
```

Output:

Result Grid

net_revenue
124714086.32

Result 6

Output

#	Time	Action	Message
19	12:25:27	WITH r AS (SELECT CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS yq, ROUND(SUM(quantity * vehicle_price * (1 - discount/100)), 2) AS net_revenue FROM order_t LIMIT 0, 50000	4 row(s) returned
20	12:27:10	SELECT ROUND(SUM(quantity * vehicle_price * (1 - discount/100)), 2) AS net_revenue FROM order_t LIMIT 0, 50000	1 row(s) returned

Result Grid

year_quarter	net_revenue	prev_revenue	qoq_change_pct
2018-Q1	39421580.16	NULL	NULL
2018-Q2	32715830.34	39421580.16	-17.01
2018-Q3	29229896.19	32715830.34	-10.66
2018-Q4	23346779.63	29229896.19	-20.13

Result 14

Output

#	Time	Action	Message
50	21:05:52	WITH revenue_by_quarter AS (SELECT CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter, ROUND(SUM(quantity * vehicle_price * (1 - discount/100)), 2) AS net_revenue FROM order_t LIMIT 0, 50000	4 row(s) returned

Observations and Insights:

- The Net Revenue is \$124,714,086.32
- Declining revenue trend every quarter in 2018, The steepest decline occurred in **Q4 (-20.13%)**.

- From Q1 to Q4, net revenue dropped by **40.8%** overall.
- The company should investigate causes of the decreased trend, continuous decline suggests deeper issues in customer satisfaction, marketing effectiveness, or product pricing.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
WITH orders_revenue_by_quarter AS (
    SELECT
        CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter,
        COUNT(*) AS total_orders,
        ROUND(SUM(quantity * vehicle_price * (1 - discount / 100)), 2) AS net_revenue
    FROM order_t
    GROUP BY year_quarter
)
SELECT
    year_quarter,
    total_orders,
    ROUND(
        ((total_orders - LAG(total_orders) OVER (ORDER BY year_quarter))
        / LAG(total_orders) OVER (ORDER BY year_quarter)) * 100, 2
    ) AS orders_qoq_pct,
    net_revenue,
    ROUND(
        ((net_revenue - LAG(net_revenue) OVER (ORDER BY year_quarter))
        / LAG(net_revenue) OVER (ORDER BY year_quarter)) * 100, 2
    ) AS revenue_qoq_pct
FROM orders_revenue_by_quarter
ORDER BY year_quarter;
```

Output:

Result Grid					
Filter Rows:		Export:		Wrap Cell Content:	
year_quarter	total_orders	orders_qoq_pct	net_revenue	revenue_qoq_pct	
2018-Q1	310	NULL	39421580.16	NULL	
2018-Q2	262	-15.48	32715830.34	-17.01	
2018-Q3	229	-12.60	29229896.19	-10.66	
2018-Q4	199	-13.10	23346779.63	-20.13	

Result 17 ×

Output

Action Output

#	Time	Action	Message
53	21:07:17	WITH orders_revenue_by_quarter AS (SELECT CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter, COUNT(*) ...	4 row(s) returned

Observations and Insights:

- The total number of orders decreased from **310 in Q1** to **199 in Q4**. Net revenue followed the same downward trend from **\$39.4M** to **\$23.3M**
- There is a consistent declining trend for both revenue and order count across quarters, confirming a loss in customer engagement over time.
- The company may need to focus on retention strategies, improve product quality or customer experience, and review pricing policies.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount),2) AS avg_credit_card_disc
FROM order_t o
    JOIN customer_t c
    ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_credit_card_disc DESC;
```

Note: The 'discount' field in the 'order_t' table is stored as a percentage, i.e. 0.6 represents a discount of 0.6%, not 60%.

Output:

credit_card_type	avg_credit_card_disc
laser	0.64
mastercard	0.63
visa-electron	0.62
china-unionpay	0.62
americanexpress	0.62
maestro	0.62
instapayment	0.62
jcb	0.61
switch	0.61
diners-club-carte-blanche	0.61
bankcard	0.61
diners-club-us-ca	0.61
diners-club-enroute	0.60
visa	0.60
solo	0.59
diners-club-international	0.58

Result 23 x

Output

Action Output

#	Time	Action	Message
27	15:19:58	SELECT c.credit_card_type, ROUND(AVG(o.discount),2) AS avg_credit_card_disc FROM order_t o JOIN customer_t c ON o.customer_id = c.c...	16 row(s) returned

Observations and Insights:

- Highest discounts (0.63-0.64%) Laser and Mastercard.
- Lowest discount (0.58%) Diners Club, holders receive the smallest discounts.
- The average discount offered across all credit card types is approximately 0.60% with minor variations among card providers. Expanding targeted promotion could increase sales volume and customer satisfaction across lower discount cardholders.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
    CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter,
    ROUND(AVG(DATEDIFF(ship_date, order_date)), 0) AS average_shipping_time
FROM order_t
GROUP BY year_quarter
ORDER BY year_quarter;
```

Output:

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
year_quarter	average_shipping_time			
2018-Q1	57			
2018-Q2	71			
2018-Q3	118			
2018-Q4	174			

Result 20 ×

Output

Action Output

#	Time	Action	Message
56	21:08:14	SELECT CONCAT(YEAR(order_date), '-Q', QUARTER(order_date)) AS year_quarter, ROUND(AVG(DATEDIFF(ship_date, order_date)), 0) AS aver...	4 row(s) returned

- Steady increase in shipping time, average days increased from **57 Q1** to **174 Days Q4** a continued increase every quarter suggests logistics inefficiencies and a likely cause for customer dissatisfaction.
- Possible causes, insufficient carrier capacity or reduced warehouse efficiency, Seasonal demand spike (e.g. Q4 holiday season)

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
\$124,714,086.32	1,000	994	3.14
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
\$23,346,779.63	199	97.96	21.5%

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

Business Recommendations

- Improve Shipping Efficiency and Reduce Delivery Time, the average days to ship 98 is probably causing dissatisfaction,
Action: implement logistics automation, new shipping partners and track real time delivery with dashboards.
Goal: Bring the average shipping time below 60days within two quarters.
- Enhance Customer Experience and Feedback Loop.
Action: Launch post-sale support, proactive communications for delays.
Goal: Raise average rating from 3.14 to 3.8 and food feedback above 35% next quarter.
- Focus on Core Markets and Top Vehicle Markers
Action: Prioritize marketing and inventory for high-performing brands and states.
Goal: Boost Q1 order volume by 10–15% from these regions through localized promotions.
- Launch Quarterly Review Dashboard
Action: Create a Dashboard tracking Revenue, Orders, Ratings, Shipping Time, and Feedback %.
Goal: Enable real time insights for leadership decision making and early risk detection.